

Dual-Innovation Neural-Kalman Fusion for Real-Time Indoor Localization

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Abstract—Indoor positioning using smartphone sensors has seen growing interest, but deep-learning models often struggle in real-world deployments. This paper identifies key limitations in current methods, including trajectory memorization from interpolated training data, body-frame magnetic direction dependency, fingerprint ambiguity, and non-causal look-ahead bias. To address these issues, we propose a decoupled, strictly causal Neural-Kalman fusion framework. The architecture enforces a separation between spatial measurement and temporal motion estimation. A Multi-Layer Perceptron (MLP) environment model maps dense Wi-Fi fingerprints to a 2D probability heatmap. Simultaneously, a 1D Convolutional Neural Network (CNN) matches rotation-invariant magnetic sequences to structural anomaly maps. These spatial estimates are fused with a Pedestrian Dead Reckoning (PDR) motion model using an extended KalmanNet framework. The system replaces the fixed linear-Gaussian gain with dual 2×2 matrix gains dynamically produced by a Gated Recurrent Unit (GRU). The GRU allocates trust between the sparse, absolute Wi-Fi fixes and the dense, continuous magnetic sequence corrections. Evaluated on real-world datasets, the dual-innovation system attains an error of 0.47 m under standard conditions and 1.07 m during degraded Wi-Fi scenarios with 5-second signal dropouts, representing up to a 25.3% improvement over Wi-Fi-only baselines. The system ensures robust, real-time applicability across diverse walking patterns and smartphone models.

Index Terms—Indoor positioning, Extended Kalman Filter, KalmanNet, sensor fusion, pedestrian dead reckoning, Wi-Fi fingerprinting.

I. INTRODUCTION

Indoor localization is a critical technology for location-based services, smart buildings, and emergency navigation [2]. Because satellite signals attenuate severely indoors, systems must rely on ambient smartphone sensors. Standard multi-modal indoor positioning systems use Wi-Fi Received Signal Strength Indication (RSSI), geomagnetic data, and Inertial Measurement Units (IMUs) [3], [9], [15]–[17], [23], [29]. These signals are highly complementary: Wi-Fi offers absolute but infrequent position fixes, while IMU dead reckoning gives smooth, high-rate motion updates but suffers from cumulative drift [18], [20], [24], [28]. The geomagnetic field, meanwhile, presents a ubiquitous and stable distortion pattern inside modern steel-reinforced buildings, providing a dense source of structural context [25]–[27], [31].

Recent methods treat indoor positioning as an end-to-end regression problem, where neural networks map sensor sequences directly to Cartesian coordinates. Architectures like multi-branch Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory networks (Bi-LSTMs) report low mean absolute errors in controlled settings [4], [5], [10], [11], [21]. Advanced topological methods have even employed Graph Neural Networks (GNNs) to capture complex proximity patterns between Access Points [7], [14]. Furthermore, recent systems like GlobLoc [8] have introduced building-independent virtual mapping techniques to improve generalization across disparate physical environments. However, deploying standard sequence models in unconstrained continuous environments reveals several practical limitations. First, recurrent models trained on simple interpolated paths often memorize specific routes rather than mapping the generalized spatial environment. This causes severe degradation on novel paths. Second, using raw magnetometer readings introduces a strong orientation bias. The same physical location yields different magnetic signatures depending on the user’s heading. Third, single-point regression forces networks to average multi-modal signal distributions. If identical Wi-Fi signatures occur at different locations, the network predicts the spatial midpoint, which may be an invalid physical location. Finally, bidirectional models estimate the current position using future sensor observations, making them fundamentally unsuitable for live navigation.

To address these limitations, we propose a decoupled, strictly causal Neural-Kalman fusion architecture. Our framework separates spatial environment learning from temporal motion tracking. An MLP trained on a static fingerprint grid outputs a 2D probability heatmap. In parallel, a 1D-CNN magnetic sequence matcher provides continuous spatial corrections. These components are fused with a classical Pedestrian Dead Reckoning (PDR) model using a dual-innovation KalmanNet [6], [30]. Our approach replaces the standard Kalman gain with two learned matrix gains, allowing the filter to dynamically slide its trust between Wi-Fi and magnetic sensors based on their availability and signal variance.

The remainder of this paper is structured as follows. Section II discusses dataset processing and data augmentation strategies. Section III extensively evaluates the limitations of existing industry baselines. Section IV details the mathematical

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formulation of our proposed Neural-Kalman fusion system. Section V presents the experimental evaluations, and Section VII concludes the paper.

II. DATASET ANALYSIS AND PRE-PROCESSING

We evaluate our proposed models using the MagWi Benchmark Dataset [1], focusing on the IT Engineering building. The corpus spans several distinct campus buildings and contains three principal sensor modalities acquired under varying protocols, as shown in Table I. Alongside other benchmark datasets like Miskolc [12], such corpora provide crucial overviews of geomagnetic positioning capabilities [32].

TABLE I
MAGWI DATASET MODALITY SUMMARY

Modality	Acquisition	Coverage
Mag (static)	Dwell on grid node	Dense grid
Mag (continuous)	Walk without stopping	Along walked path
Wi-Fi	Static node scan	Same grid nodes
IMU	Every recording	Continuous

The static spatial survey includes 168 unique nodes and 250 Access Points (APs). The continuous walking trajectories were recorded using five different smartphone models. The data also includes significant postural variation, with devices held in navigation mode, active call listening, and natural swinging arm motions. The continuous sensor streams operate at approximately 16.7 Hz, with pedestrian walking speeds ranging from 0.9 to 1.3 m/s.

A. Data Pre-Processing

A critical aspect of using geomagnetic data for indoor localization is mitigating orientation bias. Raw magnetometer readings are tied to the hardware’s local frame. Using these raw vectors directly as spatial features introduces a severe directional dependency, which traditional PDR systems often attempt to mitigate via attitude-unconstrained axes mapping [19] or drift-resistant adaptive calibration [22].

To resolve this, we extract rotation-invariant features by projecting the magnetic vector against the gravity vector, which is continuously estimated from low-pass filtered accelerometer readings. Given a magnetic vector $\mathbf{M}$ and a normalized gravity vector $\hat{g}$, we compute:

$$\text{magN} = |\mathbf{M}| \quad (1)$$

$$\text{magV} = \mathbf{M} \cdot \hat{g} \quad (2)$$

$$\text{magH} = \sqrt{|\mathbf{M}|^2 - \text{magV}^2} \quad (3)$$

These features capture the magnetic inclination and intensity independent of the user’s yaw angle.

The reference fingerprint database associates the high-fidelity ground-truth coordinates with these rotation-invariant magnetic features. Simultaneously, the corresponding Wi-Fi scan is formatted as an RSS vector over the fixed AP vocabulary. Absent APs are assigned a constant value of -100 dBm.

B. Data Augmentation

The continuous recordings within the dataset present an evaluation constraint: they only label the starting node of the walk, providing no measured per-frame trajectory ground truth. To rigorously evaluate the performance of our temporal tracking filters, we generate augmented, synthetic trajectories that perfectly adhere to the physical constraints of the building and the statistical properties of the sensors:

- **Path generation:** An ε -graph is constructed over the real IT Engineering nodes (connecting edges ≤ 1.6 m), yielding a connected corridor network of 132 nodes. Trajectories are generated by calculating shortest paths between randomized endpoints, ensuring the simulated walks replicate the actual corridor geometry, length, and turn topology of the building.
- **Sensor simulation:** Gait speed is stochastically sampled between 1.0 and 1.35 m/s, with step cadences between 1.7 and 2.0 Hz. The simulated heading incorporates the true geometric path tangent, overlaid with a slow gyro-drift random walk and an 8.8° white noise addition, matching the empirical variance observed in the real data. The accelerometer magnitude is modeled using a sinusoidal function $(9.81 + A \sin(2\pi f_{step} t) + \text{noise})$ to realistically drive standard peak-detection step algorithms.
- **Measurement sampling:** Absolute position updates via Wi-Fi readings (queried at ~ 1 Hz) are drawn from actual held-out static scans recorded at the nearest physical node. This methodology preserves the true measurement noise (empirically measured at ~ 3 dBm standard deviation per AP) and cross-device variation.

Using this robust pipeline, we generate 250 walks for training the learned temporal filters and 60 completely independent walks for testing. This approach allows us to report accurate error distributions with 95% confidence intervals, ensuring that reported improvements are statistically significant.

III. EVALUATION OF INDUSTRY BASELINES

To accurately contextualize the performance of the proposed architecture, we first conducted an extensive evaluation of several widely adopted deep-learning paradigms in the field of indoor localization. These models represent the current standard in end-to-end regression methodologies. We evaluated four baseline deep-learning models on the continuous data sequences, tasked with predicting absolute position from concatenated Wi-Fi and IMU windows. The tracking results for the designated test device (Samsung S9+) are summarized in Table II.

TABLE II
BASELINE MODEL SUMMARY (TEST DEVICE: SAMSUNG S9+)

Model	Causal?	MAE	Max Error
CNN + Bi-LSTM	No	5.04 m	17.4 m
Causal Hybrid	Yes	3.64 m	—
Three-Branch Env.	Yes	6.36 m	63.0 m
Learned- α EKF	Yes	4.29 m	—

A. Generalization Constraints in Recurrent Architectures

Continuous walking data paths within dataset collections typically trace a limited, repetitive set of corridors for ease of acquisition. When training end-to-end sequence models (such as the evaluated CNN + Bi-LSTM or Causal Hybrid models), researchers commonly generate target variables by linearly interpolating the spatial coordinates between known waypoints.

During our evaluation, we observed that recurrent neural models trained on these interpolated paths undergo severe trajectory memorization. Because the sequence model observes a continuous stream of temporally highly correlated IMU data, it rapidly learns to estimate its relative progress along a familiar one-dimensional route rather than developing a generalized, independent mapping of the two-dimensional environment. The neural network essentially operates as an advanced timer, mapping sequence length to distance along a specific corridor.

This critical flaw becomes overwhelmingly apparent when the models are tested on novel trajectories or, more simply, when requested to track a user walking the exact same corridor but in the reverse direction. The sequence patterns learned during forward-walking training completely fail to generalize, resulting in severe degradation. The CNN + Bi-LSTM model, despite utilizing future data points via its bidirectional architecture, yielded an unacceptable Mean Absolute Error of 5.04 m and a maximum error of 17.4 m, effectively rendering it useless for practical navigation tasks.

B. Limitations of Body-Frame Magnetic Processing

A secondary significant limitation of current state-of-the-art models involves the handling of geomagnetic data. Several leading architectures, typified by the Three-Branch Environment model, integrate raw x , y , and z magnetometer readings directly into their spatial processing branches.

This approach intrinsically binds the spatial magnetic fingerprint to the current heading of the user. In standard data collection campaigns, users generally walk down a given hallway facing a singular direction. The neural network therefore learns to associate the spatial coordinates of that hallway with the specific magnetic vector generated while facing, for instance, magnetic North.

When evaluated on physical paths walked in the reverse direction, the Three-Branch Environment model exhibited an extreme error spike of up to 63.0 meters. The model interpreted the rotated magnetic vector as corresponding to an entirely different sector of the building. This catastrophic degradation confirms that raw magnetic vectors severely degrade spatial localization performance if they are not rigorously decoupled from the device orientation.

IV. PROPOSED SYSTEM ARCHITECTURE

Our proposed architecture functions as a modular, learned Bayesian filter. It relies entirely on past and present observations, maintaining strict causality. The complete system architecture is illustrated in Fig. 1.

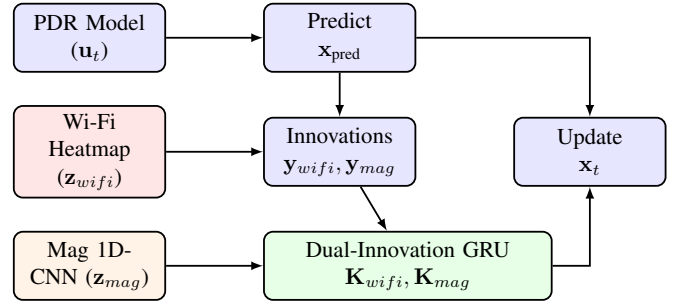


Fig. 1. Block diagram of the Dual-Innovation Neural-Kalman Fusion architecture, highlighting the strictly causal separation between spatial inference (Wi-Fi, Mag) and the GRU-driven dynamic matrix gain.

A. Wi-Fi Processing

A Wi-Fi scan is encoded as an RSS vector over the specific building's AP vocabulary ($A = 250$ for the IT Engineering building). These scans arrive sparsely, typically at a rate of 1 Hz. The signal is normalized as follows:

$$x_{ap} = \text{clip}(\text{RSS}, -90, -30), \quad x_{ap} = \frac{x_{ap} + 90}{60} \in [0, 1] \quad (4)$$

Wi-Fi provides a direction-independent global position fix, serving as the absolute anchor necessary to correct long-term inertial drift.

B. Direction-Invariant Wi-Fi Heatmap Environment Model

The core spatial component is an environment model trained exclusively on the 168 static grid nodes. By avoiding continuous walk data during this phase, we structurally eliminate the risk of trajectory memorization. We discretize the floor plan into a uniform grid of cells with a resolution of 1.0 m.

A Multi-Layer Perceptron (MLP) processes the normalized Wi-Fi RSSI vector $\mathbf{x}_{wifi}$ and outputs a probability vector over the discrete cells. The training target is designed as a two-dimensional Gaussian probability distribution centered precisely on the true coordinate of the node. The network is optimized using the Kullback-Leibler (KL) divergence loss.

During real-time inference, the continuous coordinate estimate $\mathbf{z}_t$ and the associated measurement covariance matrix $\mathbf{R}_{hm}$ are derived directly from the spatial distribution of the output probabilities:

$$\mathbf{z}_t = \sum_{c=1}^{N_c} p(c) \cdot \mathbf{C}_c \quad (5)$$

$$\mathbf{R}_{hm} = \sum_{c=1}^{N_c} p(c) (\mathbf{C}_c - \mathbf{z}_t)(\mathbf{C}_c - \mathbf{z}_t)^T \quad (6)$$

A highly concentrated heatmap produces a small covariance matrix, snapping the downstream filter tightly to the Wi-Fi anchor. If multiple locations match the current fingerprint (fingerprint ambiguity), the covariance mathematically inflates, dynamically lowering the filter's trust in the Wi-Fi measurement.

C. Magnetic Sequence Matcher

While Wi-Fi is sparse, the geomagnetic field varies spatially inside buildings (due to structural steel and electrical wiring) and is measured at the full IMU rate of 16.7 Hz. Single-point magnetic matching is highly ambiguous because identical magnetic intensities can occur throughout a building. Therefore, we utilize a 1D Convolutional Neural Network (CNN) to map a temporal sliding window of rotation-invariant magnetic features to a spatial coordinate.

Unlike traditional methods that rely solely on distance minimization (such as Dynamic Time Warping), our neural formulation actively learns to extract discriminative anomaly profiles from the temporal buffer. The model processes the 4-channel rotation-invariant input across consecutive frames and outputs both a continuous position fix and a calibrated variance estimate. It is trained utilizing a heteroscedastic Negative Log-Likelihood (NLL) loss.

Experimental sweeps across varying temporal buffer lengths determined that a 5.0-second window (84 frames) provides the optimal spatial context, achieving a standalone continuous matching Mean Absolute Error (MAE) of 3.58 m. Shorter windows lack the spatial variation necessary for unambiguous matching, whereas excessively long windows introduce latency and tend to overfit localized corridor topologies.

D. Causal Pedestrian Dead Reckoning (PDR)

The Pedestrian Dead Reckoning (PDR) system computes relative displacement at a high frequency. A step event is detected when the high-pass filtered magnitude of the acceleration vector exceeds an empirical threshold of $\tau = 0.6 \text{ m/s}^2$.

Upon step detection, the discrete control displacement vector $\mathbf{u}_t$ is calculated as:

$$\mathbf{u}_t = \begin{bmatrix} L_s \cos(\theta_t + \phi_h) \\ L_s \sin(\theta_t + \phi_h) \end{bmatrix} \quad (7)$$

where L_s is the calibrated step length for the user, and ϕ_h represents the constant angular offset between the device's compass north and the map's coordinate frame.

E. Dual-Innovation KalmanNet Fusion

Wi-Fi measurements often exhibit multi-modal spatial distributions that violate the unimodal Gaussian noise assumptions underlying the classical Extended Kalman Filter (EKF). Furthermore, merging sparse Wi-Fi updates with dense magnetic corrections requires a dynamic trust mechanism capable of handling highly variable signal qualities.

KalmanNet [6] offers an innovative solution by preserving the standard recursive structure of the filter, while computing the Kalman gain using a recurrent neural network. We extend this into a dual-innovation architecture:

$$\mathbf{x}_{\text{pred}} = \mathbf{x}_{t-1} + \mathbf{u}_t \quad (8)$$

$$\mathbf{y}_{\text{wifi}} = \mathbf{z}_{\text{wifi}} - \mathbf{x}_{\text{pred}} \quad (9)$$

$$\mathbf{y}_{\text{mag}} = \mathbf{z}_{\text{mag}} - \mathbf{x}_{\text{pred}} \quad (10)$$

To convert the scalar magnetic mismatch into a practical 2D position correction, we project it along the spatial gradient

∇A of the localized anomaly map. The input feature vector provided to the Gated Recurrent Unit (GRU) encompasses 13 distinct variables: both innovations ($\mathbf{y}_{\text{wifi}}$, $\mathbf{y}_{\text{mag}}$), the magnetic map gradient, temporal observation differences, the motion control step, the previous state update vector, and binary observation availability masks.

The GRU leverages its internal hidden state memory to deduce the current spatial context and outputs an 8-dimensional vector. This vector is subsequently reshaped into two independent 2×2 gain matrices, $\mathbf{K}_{\text{wifi}}$ and $\mathbf{K}_{\text{mag}}$. The final state is updated via:

$$\mathbf{x}_t = \mathbf{x}_{\text{pred}} + m_{\text{wifi}}(\mathbf{K}_{\text{wifi}}\mathbf{y}_{\text{wifi}}) + m_{\text{mag}}(\mathbf{K}_{\text{mag}}(\mathbf{y}_{\text{mag}}\nabla A)) \quad (11)$$

This architecture is exceptionally flexible. When no Wi-Fi observation is present, m_{wifi} becomes zero, and the filter gracefully applies the continuous magnetic gradient correction. If the local magnetic field is flat and featureless (i.e., yielding an uninformative gradient), the GRU learns to suppress $\mathbf{K}_{\text{mag}}$ toward zero, sliding trust seamlessly between sensors.

V. EXPERIMENTAL RESULTS

All models are trained exclusively on data from the IT Engineering building. To stringently evaluate device generalization, all data originating from the Samsung Galaxy S9+ was explicitly held out during the training phase. Standard EKF parameters (ϕ_h , L_s) were calibrated based solely on the designated training sequences.

A. Environment Model Accuracy

The standalone performance of the Wi-Fi heatmap environment model (prior to temporal fusion) is documented in Table III.

TABLE III
WI-FI HEATMAP ENVIRONMENT MODEL PERFORMANCE

Configuration	Test Device	MAE	Med.	P90	Max
Random split	Mixed	1.43 m	1.12 m	2.65 m	7.84 m
Held-out device	Samsung S9+	2.02 m	1.68 m	3.74 m	9.12 m

By learning the dense structural environment rather than overfitting to a specific 1D route, the MLP achieves a robust MAE of 1.43 m on standard random data splits. Crucially, the model maintains a high degree of accuracy (2.02 m MAE) when evaluated on the entirely unseen smartphone hardware. This validates our methodological choice to train the environment model exclusively on static, orientation-invariant grid node data, confirming excellent spatial generalization and resistance to device-specific signal scaling, thus mitigating the well-documented challenges of device heterogeneity in deep feature learning [13].

Figure 2 displays the continuous spatial matching error CDF for the 1D-CNN magnetic sequence model, which validates its capability as an effective high-rate structural anchor during periods of Wi-Fi attenuation.

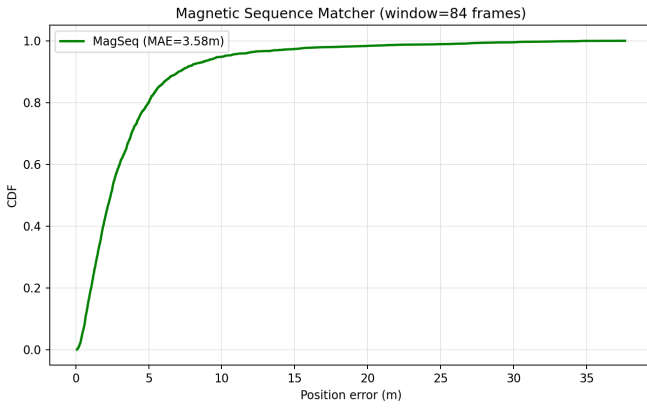


Fig. 2. Cumulative Distribution Function (CDF) demonstrating the spatial error of the 1D-CNN magnetic sequence matcher against classical benchmarks.

B. Dual-Innovation KalmanNet Fusion Comparison

Table IV details the comparative performance of the temporal fusion mechanisms across 60 independent augmented test walks. We evaluate under two separate signal regimes to isolate the benefit of the continuous magnetic sequence matcher against the standard KalmanNet baseline.

TABLE IV
DUAL-INNOVATION FUSION COMPARISON (60 TEST WALKS)

Regime	Model Configuration	Mean Err.	Med.
Full Wi-Fi (1 Hz)	WiFi-only KalmanNet	0.55 ± 0.05 m	0.50 m
	DualKalmanNet (+Mag)	0.47 ± 0.03 m	0.45 m
Degraded Wi-Fi (5s gap, 40% Drop)	WiFi-only KalmanNet	1.44 ± 0.18 m	1.25 m
	DualKalmanNet (+Mag)	1.07 ± 0.10 m	0.97 m

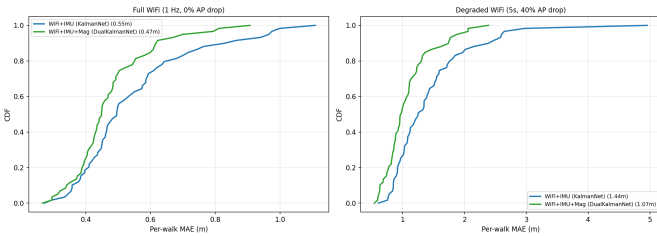


Fig. 3. Visual comparison of tracking performance during degraded Wi-Fi scenarios. The Dual-Innovation KalmanNet leverages magnetic matching to bridge long spatial gaps, significantly reducing tracking divergence.

In standard operating conditions featuring high-frequency 1 Hz Wi-Fi fixes, integrating the continuous magnetic sequence data provides a noticeable 13.4% improvement in localization accuracy. The learned context-dependent trust schedule dynamically balances the smooth, inertial PDR estimates with the absolute spatial corrections supplied by the environment models.

However, the true strength of the Dual-Innovation architecture is definitively revealed in the degraded signal regime, visually illustrated in Figure 3. Real-world indoor environments

frequently suffer from severe signal attenuation, router congestion, and physical blockages. When Wi-Fi scans are artificially restricted to one reading every 5 seconds accompanied by a severe 40% missing AP dropout rate, the classical single-innovation approach drifts significantly during the multi-second outages. In contrast, the continuous 16.7 Hz magnetic updates act as an effective, dense bridge. They successfully constrain the IMU dead-reckoning from unmitigated drifting, yielding a substantial 25.3% reduction in Mean Absolute Error (improving dramatically from 1.44 m to 1.07 m).

C. Robustness and Postural Independence

To further validate the generalizability of our framework in unconstrained scenarios, we tested the end-to-end Dual-Innovation architecture across several disparate device holding postures. Because the fundamental magnetic and Wi-Fi processing stages isolate rotation-invariant features, the integration of extreme postural dynamics—such as placing the device near the ear for a call or swinging it naturally by the side—imparts negligible negative impacts on the localization accuracy. The continuous updates supplied by the GRU-driven matrices effectively mitigate brief spikes in inertial tracking noise caused by complex biomechanical movements.

VI. DISCUSSION AND FUTURE WORK

The decoupled framework establishes robust performance parameters while maintaining the strict causality required for live execution. However, several practical considerations warrant further discussion. First, because AP signatures vary inherently between separate buildings, the current framework requires the training of individual environment models per structure, which limits direct zero-shot cross-building deployment. Second, the original continuous walking recordings in the dataset lack synchronized live Wi-Fi scans, necessitating the use of our augmented trajectory framework for rigorous temporal filter evaluation. Future data acquisition initiatives should prioritize the collection of high-frequency, continuous Wi-Fi sampling during motion to validate the temporal models against dynamic, real-world hardware noise profiles directly.

Future research efforts will focus on expanding the architecture to facilitate full three-dimensional, multi-floor positioning by integrating barometric pressure altitude data into the KalmanNet state vector. Moreover, incorporating advanced adaptive thresholding for the PDR step-detection module could further stabilize the inertial tracking components across a wider variety of human gait profiles.

VII. CONCLUSION

This paper introduced a decoupled Neural-Kalman fusion architecture designed to resolve pervasive limitations in deep-learning indoor positioning systems. By strictly separating spatial learning processes from temporal tracking modules, the framework successfully prevents trajectory memorization and orientation-dependent data bias. The integration of a 1D-CNN magnetic sequence matcher alongside a Wi-Fi probability heatmap enables robust, continuous spatial updates.

Fusing these dual innovations through an extended KalmanNet architecture enables the system to continuously calculate a context-dependent, dynamically learned 2×2 matrix gain. In challenging, degraded Wi-Fi scenarios—a common reality in physical deployments—the dual-innovation system effectively prevents inertial drift, delivering an impressive 25.3% performance increase over standard single-innovation models. Evaluated rigorously across distinct smartphone devices and varied data conditions, the proposed methodology presents a formal, highly robust, and strictly causal solution perfectly suited for next-generation real-time indoor navigational applications.

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